MMDU MULLANA OFFICIAL SYLLABUS • BMAT-001 • UNIT 3

Unit 3: Unit III: Multivariable Calculus (Differentiation) (10 Hours)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 35 mins

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1. Limits, Continuity & Partial Derivatives

1. Multivariable Functions: Limits, Continuity & Partial Derivatives

A. Limit of a Function of Two Variables:

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous ϵ-δ Definition: We write $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

- **Path Independence Principle:** For a limit to exist in R^2, the function must approach the **exact same value** L **along EVERY conceivable path** approaching (a, b).

- **Two-Path Non-Existence Test:** If $f(x,y)$ yields different values along paths $y = mx$ or $y = mx^2$ (i.e., limit depends on m), the limit **strictly DOES NOT EXIST**!

B. Continuity of $f(x,y)$:

A function $f(x,y)$ is continuous at (a,b) if:

1. $f(a,b)$ is defined.

2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.

3. $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

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2. Partial Derivatives & Higher-Order Derivatives

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \mathbf{f_x(x,y)} = \lim_{\mathbf{\Delta x \to 0}} \frac{\mathbf{f(x+\Delta x, y) - f(x,y)}}{\mathbf{\Delta x}}, \quad \frac{\partial \mathbf{f}}{\partial \mathbf{y}} = \mathbf{f_y(x,y)} = \lim_{\mathbf{\Delta y \to 0}} \frac{\mathbf{f(x,y+\Delta y) - f(x,y)}}{\mathbf{\Delta y}}$$

✅ **FUNDAMENTAL LAW / RULE:**
Clairaut's Symmetry Theorem: If the mixed partial derivatives f_{xy} and f_{yx} are both **continuous** on an open region containing (a,b), then:

$$\frac{\partial^2 \mathbf{f}}{\partial \mathbf{y} \partial \mathbf{x}} = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{x} \partial \mathbf{y}} \iff \mathbf{f_{yx} = f_{xy}}$$

University Question (4 Marks):
Evaluate whether $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ exists.

Step 1 🚀: Approach along straight line paths $y = mx$

$$\lim_{x\to 0} \frac{x^2(mx)}{x^4 + m^2x^2} = \lim_{x\to 0} \frac{mx}{x^2 + m^2} = 0$$

Step 2 ⚡: Approach along parabolic path $y = mx^2$

$$\lim_{x\to 0} \frac{x^2(mx^2)}{x^4 + m^2x^4} = \frac{m}{1 + m^2}$$

Since the limit depends on m, the limit is **not unique** and **does NOT exist**.

🎯 By Path Test ($y = mx^2$), the limit **does NOT exist**.

University Question (6 Marks - MMDU Question Paper):
If $u = \ln(x^2 + y^2)$, prove that u satisfies 2D Laplace's Equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Step 1 🚀: First Partial Derivatives

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}$$

Step 2 ⚡: Second Partial Derivatives via Quotient Rule

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(x^2 + y^2) - 2x(2x)}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$$

Step 3 🎯: Sum the Second Derivatives

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2y^2 - 2x^2 + 2x^2 - 2y^2}{(x^2 + y^2)^2} = 0$$

🎯 Proved: $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (u is Harmonic).

2. Euler's Theorem on Homogeneous Functions

1. Homogeneous Functions & Euler's Theorem: Rigorous Analysis

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A multivariable function $f(x,y)$ is **homogeneous of degree** n if for any scalar $t > 0$:

$$\mathbf{f(tx, ty) = t^n f(x, y) \iff f(x, y) = x^n \phi\left(\frac{y}{x}\right)}$$

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2. Euler's Theorem on Homogeneous Functions (First & Second Order)

✅ **FUNDAMENTAL LAW / RULE:**
First-Order Euler's Theorem: If $u = f(x,y)$ is a homogeneous function of x and y of degree n with continuous partial derivatives, then:

$$\mathbf{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u}$$

✅ **FUNDAMENTAL LAW / RULE:**
Second-Order Euler's Formula:

$$\mathbf{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u}$$

🚨 **EXAMINER TRAP / PITFALL:**
Examiner's Composite Function Deduction: If $z = f(u)$ is homogeneous of degree n in (x,y), then:

1. $xu_x + yu_y = n\frac{f(u)}{f'(u)} = g(u)$

> 2. $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$

University Question (6 Marks - MMDU Question Paper):

If $u = \sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$, prove that $xu_x + yu_y + zu_z = -3\tan u$.

Step 1 🚀: Convert to Homogeneous Function $v = \sin u$

$$v = \sin u = \frac{x + 2y + 3z}{\sqrt{x^8 + y^8 + z^8}}$$

Degree of numerator is 1; degree of denominator is $\sqrt{t^8} = t^4$. Thus v is homogeneous of degree $n = 1 - 4 = -3$.

Step 2 ⚡: Apply 3-Variable Euler's Theorem to v

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} + z\frac{\partial v}{\partial z} = -3v$$

Using $\frac{\partial v}{\partial x} = \cos u \frac{\partial u}{\partial x}$:

$$\cos u\,(xu_x + yu_y + zu_z) = -3\sin u \implies xu_x + yu_y + zu_z = -3\tan u$$

🎯 Result Proved: $\mathbf{x}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \mathbf{y}\frac{\partial \mathbf{u}}{\partial \mathbf{y}} + \mathbf{z}\frac{\partial \mathbf{u}}{\partial \mathbf{z}} = -\mathbf{3}\tan \mathbf{u}$.

University Question (6 Marks - MMDU Previous Paper):

If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$.

Step 1 🚀: Define $z = \tan u$ **(Degree** $n = 3 - 1 = 2$**)**

$$xz_x + yz_y = 2z \implies x\sec^2 u\,u_x + y\sec^2 u\,u_y = 2\tan u \implies xu_x + yu_y = \frac{2\tan u}{\sec^2 u} = \sin 2u = g(u)$$

Step 2 ⚡: Second-Order Euler Identity

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1] = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$$

🎯 Second-Order Euler Identity Proved: $\mathbf{x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \sin 2u(2\cos 2u - 1)}$.

3. Total Derivative, Maxima-Minima & Lagrange Multipliers

1. Total Differential, Chain Rule & Extrema of Multivariable Functions

A. The Total Differential (du) & Total Derivative ($\frac{du}{dt}$):

- **Total Differential:** $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$

- **Total Derivative (Single Parameter t):**

$$\frac{\mathbf{du}}{\mathbf{dt}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\mathbf{dx}}{\mathbf{dt}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\mathbf{dy}}{\mathbf{dt}}$$

- **Composite Chain Rule ($x = x(r,\theta),\ y = y(r,\theta)$):**

$$\frac{\partial \mathbf{u}}{\partial \mathbf{r}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \mathbf{r}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \mathbf{r}}, \quad \frac{\partial \mathbf{u}}{\partial \theta} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \theta} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \theta}$$

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2. Unconstrained Maxima & Minima of Two Variables ($rt - s^2$ Test)

To locate the local maxima, local minima, and saddle points of $z = f(x,y)$:

1. **Stationary Points:** Solve $f_x = 0$ and $f_y = 0$.

2. **Second-Order Partials at (a,b):** $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$ and Discriminant $\mathbf{\Delta = rt - s^2}$:

Condition on $\Delta = rt - s^2$	Condition on $r = f_{xx}$	Nature of Critical Point (a,b)
$\mathbf{rt - s^2 > 0}$	$\mathbf{r < 0}$	**Local Maximum**
$\mathbf{rt - s^2 > 0}$	$\mathbf{r > 0}$	**Local Minimum**
$\mathbf{rt - s^2 < 0}$	Any sign	**Saddle Point**
$\mathbf{rt - s^2 = 0}$	Indeterminate	**Test Fails** (Further investigation required)

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3. Constrained Optimization: Method of Lagrange Multipliers

To optimize $f(x, y, z)$ subject to constraint $\phi(x, y, z) = 0$:

1. Form Auxiliary Lagrangian: $\mathbf{F(x, y, z, \lambda) = f(x, y, z) + \lambda\, \phi(x, y, z)}$.

2. Solve the stationary system:

$$\mathbf{f_x + \lambda\phi_x = 0}, \quad \mathbf{f_y + \lambda\phi_y = 0}, \quad \mathbf{f_z + \lambda\phi_z = 0}, \quad \mathbf{\phi(x, y, z) = 0}$$

University Question (4 Marks):
Find the local maximum and minimum values of $f(x, y) = x^3 + y^3 - 3xy$.

Step 1 🚀: Find Critical (Stationary) Points
$f_x = 3x^2 - 3y = 0 \implies y = x^2$ and $f_y = 3y^2 - 3x = 0 \implies 3x(x^3 - 1) = 0$.
Critical points: $(0, 0)$ and $(1, 1)$.

Step 2 ⚡: Compute Second Partials r, s, t
$r = f_{xx} = 6x, \quad s = f_{xy} = -3, \quad t = f_{yy} = 6y \implies \Delta = rt - s^2 = 36xy - 9$.

Step 3 🎯: Classify Points

1. **At** $(0, 0)$**:** $\Delta = 0 - 9 = -9 < 0 \implies (0, 0)$ is a **Saddle Point**.

2. **At** $(1, 1)$**:** $\Delta = 36 - 9 = 27 > 0$ and $r = 6 > 0 \implies (1, 1)$ is a **Local Minimum** with $f_{\min}(1, 1) = 1 + 1 - 3 = -1$.

🎯 Saddle point: $(0, 0)$ | Local Minimum at $(1, 1)$ with $\mathbf{f_{min} = -1}$.

University Question (6 Marks - MMDU Question Paper):
Find the maximum volume of a rectangular box open at the top, having a given surface area $S = 108 \text{cm}^2$, using Lagrange's method of multipliers.

Step 1 🚀: Objective and Constraint Functions
Maximize $V = xyz$ subject to $\phi(x, y, z) = xy + 2yz + 2zx - 108 = 0$.
Lagrangian: $F(x, y, z, \lambda) = xyz + \lambda(xy + 2yz + 2zx - 108)$.

Step 2 ⚡: Set First Partials to Zero

$$yz + \lambda(y + 2z) = 0, \quad xz + \lambda(x + 2z) = 0, \quad xy + \lambda(2x + 2y) = 0$$

Equating λ gives $x = y = 2z \implies z = \frac{x}{2}$.

Step 3 🎯: Substitute into Constraint $S = 108$

$$x^2 + 2x(x/2) + 2x(x/2) = 3x^2 = 108 \implies x = 6,\ y = 6,\ z = 3 \implies V_{\max} = 6 \times 6 \times 3 = 108 \text{cm}^3$$

🎯 Dimensions: 6cm × 6cm × 3cm | Maximum Volume: $\mathbf{V_{max} = 108 cm^3}$.

4. Scalar & Vector Functions, Gradient, Divergence & Curl

1. Vector Differential Calculus: Del, Gradient, Divergence & Curl

✅ **FUNDAMENTAL LAW / RULE:**
The Vector Differential Operator Del (∇):

$$\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$$

A. Gradient of a Scalar Field ($\nabla\phi$) & Directional Derivative:

- **Gradient:** $\nabla\phi = \hat{\mathbf{i}}\frac{\partial\phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial\phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial\phi}{\partial z}$ (normal to level surface $\phi(x, y, z) = c$).
- **Unit Normal Vector:** $\hat{\mathbf{n}} = \frac{\nabla\phi}{|\nabla\phi|}$.
- **Directional Derivative along $\vec{a}$:** D.D. $= \nabla\phi \cdot \frac{\vec{\mathbf{a}}}{|\vec{\mathbf{a}}|}$ (maximum magnitude = $|\nabla\phi|$ in direction of $\nabla\phi$).

B. Divergence ($\nabla \cdot \vec{F}$) & Curl ($\nabla \times \vec{F}$):

- **Divergence (Scalar):** $\text{div}\,\vec{\mathbf{F}} = \nabla \cdot \vec{\mathbf{F}} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$.

- **Solenoidal Field:** $\nabla \cdot \vec{\mathbf{F}} = 0$.

- **Curl (Vector):** $\text{curl}\,\vec{\mathbf{F}} = \nabla \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$.

- **Irrotational (Conservative) Field:** $\nabla \times \vec{\mathbf{F}} = \vec{0} \iff \vec{\mathbf{F}} = \nabla\phi$.

University Question (6 Marks - MMDU Question Paper):
Find the directional derivative of $\phi = x^2yz + 4xz^2$ at point $P(1,-2,-1)$ in the direction of vector $\vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Step 1 🚀: Calculate $\nabla\phi$ at $P(1,-2,-1)$

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\Big|_{(1,-2,-1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Step 2 ⚡: Project onto Unit Vector $\hat{a}$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{4+1+4}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \implies \text{D.D.} = \nabla\phi \cdot \hat{a} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

🎯 Directional Derivative: $\frac{37}{3} \approx \mathbf{12.33}$.

University Question (6 Marks - MMDU Question Paper):
Prove that $\vec{F} = (y^2\cos x + z^3)\hat{i} + (2y\sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is irrotational, and find its scalar potential ϕ.

Step 1 🚀: Evaluate $\text{curl}\,\vec{F} = \nabla \times \vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2\cos x + z^3 & 2y\sin x - 4 & 3xz^2 + 2 \end{vmatrix} = (0)\hat{i} - (3z^2 - 3z^2)\hat{j} + (2y\cos x - 2y\cos x)\hat{k} = \vec{0}$$

Step 2 ⚡: Integrate Exact Differential $d\phi = \vec{F} \cdot d\vec{r}$

$$d\phi = d(y^2\sin x) + d(xz^3) - d(4y) + d(2z) \implies \phi = y^2\sin x + xz^3 - 4y + 2z + C$$

🎯 $\nabla \times \vec{F} = \vec{0}$ (Irrotational) | Scalar Potential: $\phi = \mathbf{y^2\sin x + xz^3 - 4y + 2z + C}$.

University Solved Examination Questions (Unit 3)

MMDU Official University Solved Examination Questions (Unit 3)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 7 (1 Mark): Homogeneous Degree via Euler's Theorem

Question:
Find the degree of homogeneity for $u(x,y) = \frac{x^3+y^3}{x-y}$ and evaluate $xu_x + yu_y$.

Step-by-Step Solution:
Since $u(tx,ty) = \frac{t^3(x^3+y^3)}{t(x-y)} = t^2u(x,y)$, the degree is $n = 2$. By Euler's theorem, $xu_x + yu_y = 2u$.

🎯 **Verified Result:** Degree $n = 2$ | $xu_x + yu_y = 2u$.

📘 Question 19 (4 Marks): Two-Variable Extrema Determination

Question:
Find and classify all local extrema and saddle points of $f(x,y) = x^3 + y^3 - 3axy$ $(a > 0)$.

Step-by-Step Solution:
Stationary system: $f_x = 3x^2 - 3ay = 0$ and $f_y = 3y^2 - 3ax = 0 \implies$ critical points $(0,0)$ and (a,a).
With $r = 6x, s = -3a, t = 6y$ and $\Delta = rt - s^2 = 36xy - 9a^2$:

- At $(0,0)$: $\Delta = -9a^2 < 0 \implies$ **Saddle Point**.
- At (a,a): $\Delta = 27a^2 > 0, r = 6a > 0 \implies$ **Local Minimum** with $f_{\min} = -a^3$.

🎯 **Verified Result:** $(0,0)$ is a Saddle Point; (a,a) is a Local Minimum with value $-a^3$.

📘 Question 23 (6 Marks): Constrained Optimization via Lagrange Multipliers

Question:
Find the maximum and minimum distances from the origin to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ $(a > b > c > 0)$.

Step-by-Step Solution:
Using $F = (x^2 + y^2 + z^2) + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$, setting $F_x = F_y = F_z = 0$ yields $\lambda = -a^2, -b^2, -c^2$, corresponding to distances $d_{\max} = a$ and $d_{\min} = c$.

🎯 **Verified Result:** Maximum distance $= a$ | Minimum distance $= c$.